

VEGGLI II, Norway

WMO: 014710

Lat: 60.0435N

Lon: 9.1468E

Elev: 275

StdP: 98.06

Time Zone: 1.00 (EUC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.7	-15.4	-20.6	0.6	-17.1	-18.4	0.8	-14.5	6.8	3.9	5.8	3.3	1.7	280	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	26.7	15.8	24.4	15.0	22.2	14.2	17.2	22.9	16.4	21.6	15.4	20.2	2.3	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.4	11.3	18.3	14.4	10.6	17.7	13.5	10.0	16.8	49.5	23.1	46.8	22.0	44.1	20.1	21.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 5.6	4.6	3.9		-19.6	29.7	2.9	1.8	-21.7	31.0	-23.4	32.1	-25.0	33.1	-27.1	34.4	
(5)			DB	-20.0	18.9	2.8	1.3	-22.0	19.9	-23.7	20.6	-25.2	21.3	-27.3	22.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.2	-4.9	-3.6	0.6	5.1	9.8	13.7	16.1	14.1	10.6	4.8	-0.4	-3.7
	DBStd	8.28	5.26	5.26	4.30	3.51	4.07	3.01	2.93	2.23	2.74	3.38	3.82	5.73
	HDD10.0	2270	462	381	290	152	54	3	0	1	26	165	312	424
	HDD18.3	4799	720	614	548	397	264	142	84	132	232	419	562	682
	CDD10.0	533	0	0	1	4	49	116	188	128	43	4	0	0
	CDD18.3	21	0	0	0	0	1	4	15	1	0	0	0	0
	CDH23.3	294	0	0	0	0	29	81	177	7	1	0	0	0
	CDH26.7	52	0	0	0	0	2	13	37	0	0	0	0	0
(14) Wind	WSAvg	1.7	1.7	1.7	2.0	2.0	1.9	1.8	1.6	1.4	1.6	1.5	1.5	1.6
(15) Precipitation	PrecAvg	788	54	35	33	38	63	84	100	105	85	75	71	50
	PrecMax	1125	119	129	87	100	234	183	233	204	210	225	277	105
	PrecMin	578	10	6	0	4	9	6	18	26	25	9	7	12
	PrecStd	133	28	26	23	27	47	48	50	42	43	48	55	23
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.7	10.1	14.6	19.4	26.3	28.5	29.7	24.3	22.0	16.4	10.4	10.2
		MCWB	5.1	5.5	7.8	10.2	14.6	15.6	16.8	15.8	14.1	12.3	7.0	6.5
	2%	DB	6.0	7.8	11.5	17.1	23.3	25.8	27.6	22.5	19.6	13.7	8.5	8.0
		MCWB	3.0	3.7	6.1	9.4	13.4	14.6	16.5	15.4	13.2	9.5	5.7	4.5
	5%	DB	3.6	5.7	9.1	14.3	20.6	23.1	25.4	21.0	17.6	11.9	6.5	5.8
		MCWB	1.0	2.3	4.4	7.6	12.1	13.5	16.2	14.4	12.3	8.5	4.7	3.2
	10%	DB	1.9	3.5	7.0	11.7	18.0	20.7	22.9	19.2	15.8	10.1	4.8	3.5
		MCWB	0.0	1.1	2.8	6.4	11.0	12.6	15.3	13.5	11.3	7.2	3.1	1.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.1	6.3	8.2	11.1	15.9	16.8	18.7	17.6	15.7	12.9	8.0	6.6
		MCDB	8.2	10.1	13.9	17.5	22.5	22.8	24.7	21.6	19.2	16.1	9.1	10.0
	2%	WB	3.2	4.1	6.3	9.7	14.5	15.7	17.7	16.6	14.3	10.6	6.3	4.8
		MCDB	5.5	7.0	11.1	15.7	21.2	22.6	24.1	20.3	17.6	12.5	7.8	7.3
	5%	WB	1.6	2.6	4.9	8.2	13.1	14.6	16.9	15.6	13.2	9.1	4.9	3.4
		MCDB	3.1	5.2	8.9	13.4	18.7	20.9	23.0	19.0	16.3	11.1	6.4	5.4
	10%	WB	0.4	1.3	3.2	6.9	11.6	13.6	16.0	14.5	12.2	7.8	3.5	1.7
		MCDB	1.5	3.2	6.2	11.4	16.6	19.0	21.4	17.8	14.9	9.8	4.6	3.3
(35) Mean Daily Temperature Range	MDBR		5.7	6.8	9.0	10.2	11.2	11.0	10.5	9.9	9.3	7.2	5.3	5.8
	5% DB	MCDBR	8.1	9.4	11.5	14.6	15.3	15.6	14.7	12.5	12.3	9.9	8.1	8.6
		MCWBR	6.0	6.3	7.0	7.8	7.3	7.0	6.1	6.2	7.0	6.3	6.3	6.4
	5% WB	MCDBR	7.3	8.3	11.1	13.0	13.0	12.4	12.1	10.1	10.5	8.4	7.3	8.2
		MCWBR	5.7	5.9	6.9	7.3	6.6	6.2	5.6	5.7	6.5	6.0	6.1	6.3
(40) Clear-Sky Solar Irradiance	taub		0.233	0.274	0.306	0.346	0.349	0.341	0.360	0.352	0.320	0.304	0.253	0.205
	taud		2.195	2.283	2.278	2.294	2.426	2.477	2.439	2.452	2.530	2.490	2.282	2.091
	Ebn at Noon		570	740	825	851	879	891	862	841	816	709	540	445
	Edh at Noon		39	65	90	107	102	98	100	92	72	54	36	27
(44) All-Sky Solar Radiation	RadAvg		0.33	0.99	2.50	4.08	4.81	5.35	4.73	3.74	2.46	1.15	0.42	0.21
	RadStd		0.06	0.22	0.21	0.56	0.56	0.47	0.61	0.31	0.33	0.22	0.10	0.02

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (56 neighbors)	N/A	N/A	+1.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page